

Advancing machine learning for protein structure prediction in drug discovery

Specific Aims

Aim 1: Advance machine learning for protein structure prediction in drug discovery via a materials-focused strategy.

Aim 2: Advance machine learning for protein structure prediction in drug discovery via a materials-focused strategy.

Aim 3: Advance machine learning for protein structure prediction in drug discovery via a materials-focused strategy.

Background

This proposal addresses machine learning for protein structure prediction in drug discovery, a problem with substantial significance. Our preliminary data suggest a tractable path toward measurable improvement. We will assemble a multidisciplinary team with complementary expertise. The approach integrates established materials methods with a novel analytical pipeline.

Approach

The approach integrates established materials methods with a novel analytical pipeline. Prior work has established a partial understanding, yet key gaps remain. Rigorous, pre-registered analyses will guard against bias and support reproducibility. Our preliminary data suggest a tractable path toward measurable improvement.

Innovation

Rigorous, pre-registered analyses will guard against bias and support reproducibility.

Investigator Context

The applicant is a community nonprofit partnering with an academic core. The R1 host institution offers extensive core facilities and shared instrumentation, backed by a deep institutional record of federally funded work.

Budget Justification

Budget: personnel \$1,647,060; equipment \$292,480; travel \$69,708; indirect costs \$317,744. Total requested support is \$2,326,992 over 3 years at a R1 institution.

References

1. [1] <https://doi.org/10.1145/3292500.3330701>
2. [2] <https://doi.org/10.1126/science.1259855>

3. [3] <https://doi.org/10.1145/3292500.3330701>
4. [4] <https://doi.org/10.1021/jacs.9b02765>
5. [5] <https://doi.org/10.1289/ehp.1104477>
6. [6] <https://doi.org/10.1289/ehp.1104477>
7. [7] <https://doi.org/10.1038/nature14539>
8. [8] <https://doi.org/10.1289/ehp.1104477>